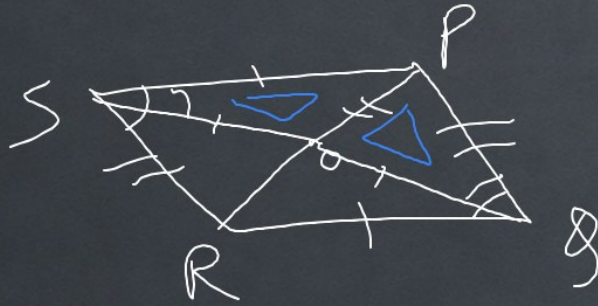




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Class: IX





Date:

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Class:

VIII

$$D = S \times T$$

$$= 60 \text{ hr}$$

$$x = 60$$

$$y = 5$$

$$k = x \times y$$

$$= 300$$

$$k = x \times y$$

$$300 = x \times 4$$

$$x = \frac{300}{4}$$

$$= 75 \text{ km/hr}$$



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Decent Coaching Classes



Class:

X

$$\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$$

$$\begin{aligned} \text{LHS} &= \tan^2 \theta (\tan^2 \theta + 1) \\ &= \tan^2 \theta \times \sec^2 \theta \end{aligned}$$

$$\begin{aligned} \text{RHS} &= \sec^4 \theta - \sec^2 \theta \\ &= \sec^2 \theta (\sec^2 \theta - 1) \\ &= \sec^2 \theta \times \tan^2 \theta \end{aligned}$$



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Decent Coaching Classes



Class:

11

$$\sec^6 x - \tan^6 x = 1 + 3 \sec^2 x \tan^2 x$$

$$\text{LHS} = (\sec^2 x)^3 - (\tan^2 x)^3$$

$$= (\sec^2 x - \tan^2 x)$$

$$\boxed{a^3 - b^3 = (a - b)(a^2 + ab + b^2)}$$

$$(\sec^2 x)^2 + \sec^2 x \tan^2 x + (\tan^2 x)^2$$

$$= (\sec^2 x)^2 + (\tan^2 x)^2 + \sec^2 x \tan^2 x$$

$$= [(\sec^2 x - \tan^2 x)^2 + 2 \sec^2 x \tan^2 x] + \sec^2 x \tan^2 x$$

$$= 1 + 3 \sec^2 x \tan^2 x$$



Date:

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Class:

X

Shivraj

100

~~999~~995

$$a = 100$$

$$d = 5$$

$$n = ?$$

$$t_n = 995$$

$$t_n = a + (n-1)d$$

$$995 = 100 + (n-1)5$$

$$179 = (n-1)$$

$$\boxed{n = 180}$$



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Decent Coaching Classes



Class:

VIII

$$\sqrt[3]{729} = 9$$

$$\sqrt[3]{0.000729}$$

$$\left(\frac{62}{3}\right)$$

$$\underline{0.09}$$



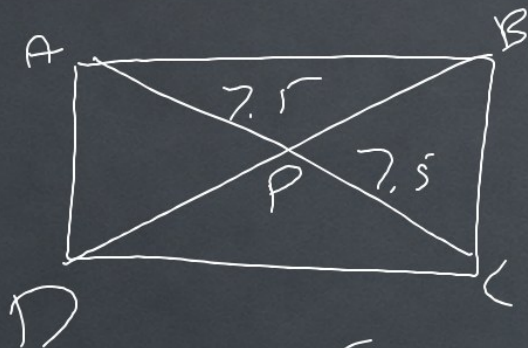
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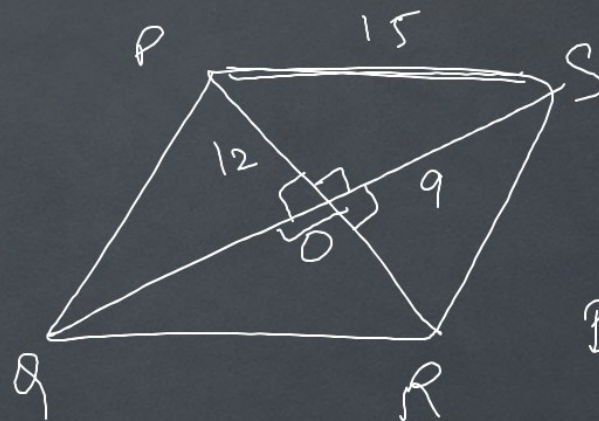
Class:

VIII



$$AC = 15$$

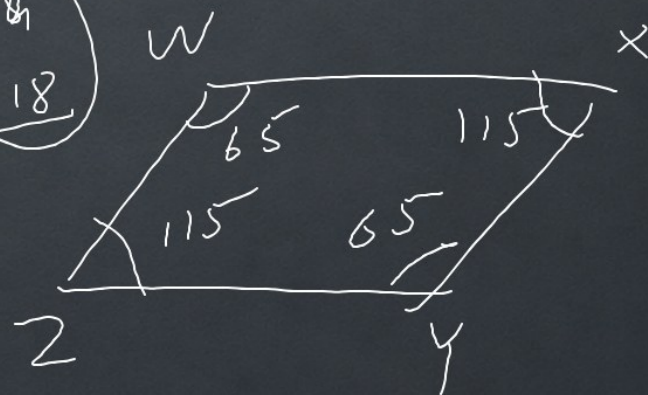
Bisect



Perp Bisect
~~Perp Bisect~~

$$PR = 24$$

$$QS = 18$$





Date:

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Class:

VII

$$y \propto \sqrt{x}$$

$$y = k \times \sqrt{x}$$

$$k = \frac{y}{\sqrt{x}}$$

$$= \frac{24}{\sqrt{16}}$$

$$= \frac{24}{4} = 6$$



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Decent Coaching Classes

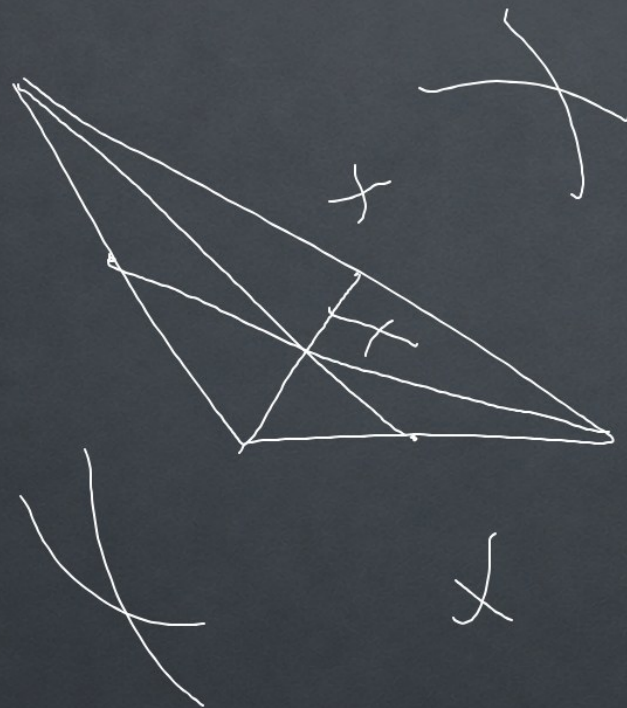


Class:

VIII

$$\begin{aligned}a &= 12 \\d &= 4 \\t_n &= 248\end{aligned}$$

4



$$P \propto \frac{1}{\sqrt{a}}$$



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Class:

VIII

$$\underline{k = xy} \text{ — Inv}$$

$$\begin{aligned} x &= \frac{k}{y} \\ y &= \frac{k}{x} \end{aligned}$$

$$\underline{k = \frac{x}{y}} \text{ — Direct}$$

$$\begin{aligned} x &= ky \\ y &= \frac{x}{k} \end{aligned}$$

$$K = \frac{x}{y}$$

$$= \frac{120}{3}$$

$$K = 40$$

$$K = \frac{x}{y}$$

$$40 = \frac{1800}{y}$$

$$\begin{aligned} y &= \frac{1800}{40} \\ &= 45 \end{aligned}$$

Date: 5/10Class: VII

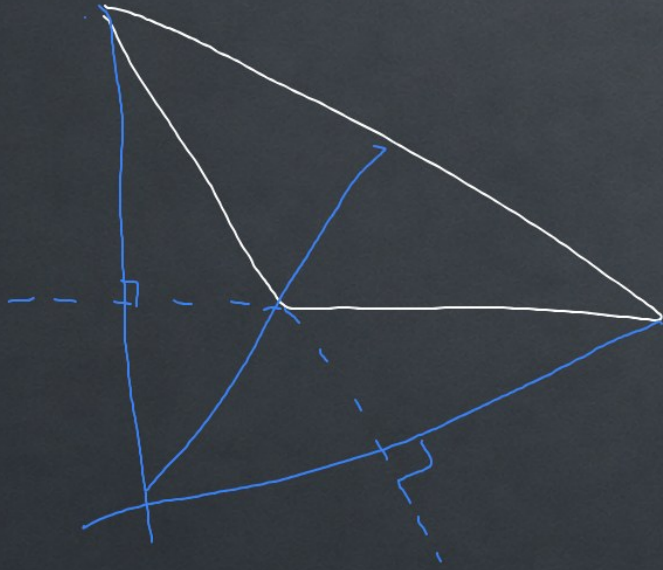
$$a + 16d = 7 + a + 9d$$

$$\underline{16d - 9d = 7}$$

$$d = 7$$

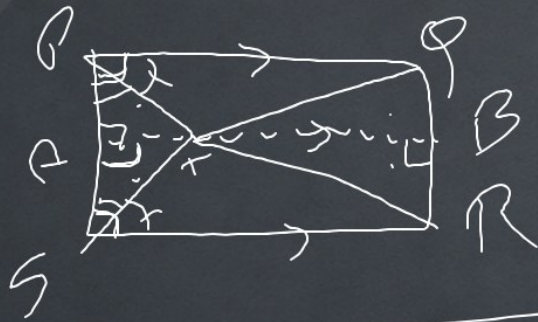
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Class: VIII
Brajesh



Date: 4/10

Class: X

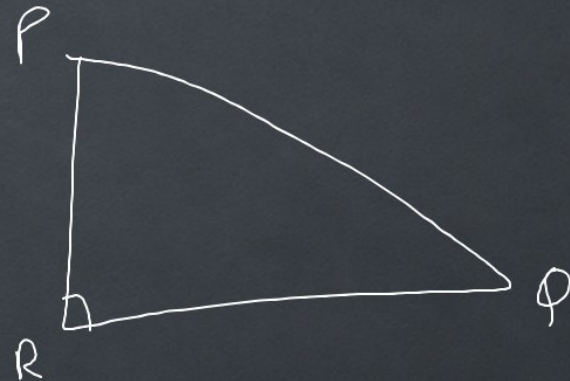


$$AT^2 + AS^2 = TS^2 \dots (1) \{PT\}$$

$$BT^2 + BQ^2 = QT^2 \dots (2)$$

$$AT^2 + AS^2 + BT^2 + BQ^2 = ST^2 + QT^2$$

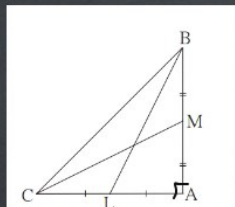
$$\frac{AT^2 + AS^2}{TP^2} + \frac{BT^2 + BQ^2}{RT^2}$$





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$$AB^2 + AC^2 = BC^2 \quad \{ \}$$

$$AB^2 + AL^2 = BL^2 \quad \{ \}$$

$$AM^2 + AC^2 = CM^2 \quad \{ \}$$

$$AB^2 + AL^2 + AM^2 + AC^2 = BL^2 + CM^2$$

$$BC^2 + \left(\frac{AC}{2}\right)^2 + \left(\frac{AB}{2}\right)^2 = BL^2 + CM^2$$



Class: _____

~~X~~

$$\frac{BC^2}{1} + \frac{BC^2}{4} = BL^2 + CM^2$$

$$\frac{5BC^2}{4} = BL^2 + CM^2$$

$$5BC^2 = 4(BL^2 + CM^2)$$



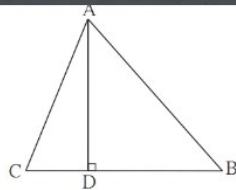
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Decent Coaching Classes

13. In $\triangle ABC$, seg $AD \perp$ seg BC $DB = 3CD$. Prove that :

$$2AB^2 = 2AC^2 + BC^2$$



Class:

X

$$AD^2 + BD^2 = AB^2 \quad \{ \}$$

$$AD^2 + CD^2 = AC^2 \quad \{ \}$$

$$AB^2 - BD^2 + CD^2 = AC^2$$

$$AB^2 - (3CD)^2 + CD^2 = AC^2$$

$$AB^2 - 9CD^2 + CD^2 = AC^2$$

$$AB^2 = AC^2 + 8CD^2$$

$$AB^2 = AC^2 + 8 \times \frac{BC^2}{16}$$

$$AD^2 = AB^2 - BD^2$$

$$DB = 3CD \quad \leftarrow$$

$$BC = CD + BD$$

$$BC = 4CD$$

$$CD = \frac{BC}{4}$$

$$CD^2 = \frac{BC^2}{16}$$

$$AB^2 = AC^2 + \frac{BC^2}{2}$$

$$\frac{2AB^2}{2} = \frac{2AC^2}{2} + \frac{BC^2}{2}$$

$$2AB^2 = 2AC^2 + BC^2$$



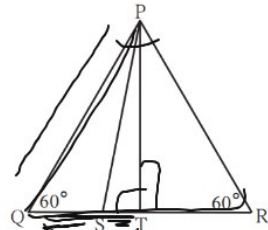
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30-60-90

Decent Coaching Classes

16*. In the figure 2.35, ΔPQR is an equilateral triangle. Point S is on seg QR such that $QS = \frac{1}{3} QR$.
Prove that : $9 PS^2 = 7 PQ^2$



Class: _____

X

$$PQ^2 - \frac{2 \times QR^2}{9 \times 4} = PS^2$$

$$PQ^2 - \frac{2 PQ^2}{9} = PS^2$$

$$\frac{7 PQ^2}{9} = PS^2$$

$$PT^2 + TQ^2 = PQ^2$$

$$PT^2 + TS^2 = PS^2$$

$$PQ^2 - TQ^2 + TS^2 = PS^2$$

$$PQ^2 - TQ^2 + \frac{1}{9} TQ^2 = PS^2$$

$$PQ^2 - \frac{9 TQ^2 + TQ^2}{9} = PS^2$$

$$PQ^2 - \frac{8 TQ^2}{9} = PS^2$$

$$QS = \frac{1}{3} \times QR$$

$$QR = 2 \times QT$$

$$QS = \frac{1}{3} \times 2 QT$$

$$= \frac{2}{3} QT$$

$$ST = \frac{1}{3} QT$$

$$ST^2 = \frac{1}{9} QT^2$$